PEST MANAGEMENT DECISION GUIDE: GREEN AND YELLOW LIST

Cassava mosaic disease in cassava - Zambia

Cassava mosaic (syn. Cassava mosaic geminiviruses (CMGs)) Ukukonkontana kwamabula na amabalamabala muli katapa in Bemba language; Kikola Kya makamba in Kaonde langua



Mosaic pattern, light yellow or green areas on the leaf blades due to cassava mosaic. (Source: CABI)



Tiny (¼ cm) whitefly is a vector of cassava mosaic. (Source: Colvin et al. 2006).



Weakened plant with curled leaves. (Source: Pasquale Piccirillo, CPC)

Prevention

- Avoid planting alternative hosts of the virus, such as castor bean adjacent to cassava field.
- Avoid planting cassava if neighbouring fields have virus-infected cassava as these could be carried by whiteflies.
- Use disease-free planting materials/ cuttings from registered suppliers, such as Golden Valley Agricultural Research Trust, Conservation Farming Unit in Chisamba, ZARI; or consult a Plant Doctor/Extension Officer.
- Avoid exchanging cuttings between farmers.
- Intercrop with maize, cowpeas or marigolds to repel whiteflies that carry the disease.
- Remove weeds to discourage habitat for the whiteflies.
- Sterilize tools used to prepare cuttings with fire or disinfectant such as sodium hypochlorite bleach (Jik), 70% ethanol (Methylated spirit).

Monitoring

- Monitor cassava plants at regular intervals to identify diseased plants and virus-transporting whiteflies.
- Examine at least 20 plants per 200 to 400 plants once a week for virus symptoms.
- Look for yellow/white mottled spots on the leaves and wrinkled/deformed leaves.
- Look for yellowing and tiny leaves, tiny stems, stunted growth and small tubers which can occur throughout the growing season.
- Threshold for action is 3-5 plants out of 20 with virus symptoms.
- When cassava is young, scout for whiteflies in early mornings or evenings once per week on 20 plants. If you shake plants, flying-off whiteflies can be seen.
- Consider control of whiteflies if 3-5 are found per plant out of 20 plants and if you have Cassava Mosaic in your area.
- Apply green direct controls as soon as disease symptoms are seen and yellow direct controls when whiteflies are present.

Direct Control

- Remove all diseased cassava or other host plants from within and immediately around sites. Dry and burn far from the field.

Direct Control

Restrictions

- Using insecticides to manage insect vectors of cassava mosaic is not advised for small-scale farmers, it is usually ineffective because whiteflies move too much.
- For larger producers growing planting materials (cuttings), it might be advisable to use insecticides to manage whiteflies.
- When using a pesticide always wear protective clothing and follow the instructions on the product label. Do not empty into drains or water sources.
- Do not apply insecticides in areas where cassava leaves are harvested for food.
- Always consult a recent list of registered pesticides e.g. from ZEMA.

• Apply cypermethrin (e.g. Cypermethryn 20EC; Cyrux 20EC) at a rate of 5ml per 20L water. Contact and stomach acting pyrethroid. IRAC group: 3A.	• WHO Class II (moderately hazardous). PHI 3-7 days. REI 3 days after spray. Max. 2 sprays per season. High risk to bees, very toxic to aquatic organisms. Do not spray near water.

Certificate of Advanced Studies- "Sustainable Crop Production"

Zambia

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